

Sireesha Borra
Senior Python Developer (AI/ML)
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PROFESSIONAL SUMMARY:

- Senior Python Backend Developer with 9+ years of experience designing, developing, and implementing scalable backend systems for enterprise applications.
- Expertise in Python web frameworks including FastAPI, Django, Flask for building modular and maintainable backend services efficiently.
- Hands-on experience with Spring Boot framework for enterprise-grade backend modules and microservices development.
- Strong track record in developing, maintaining, and optimizing RESTful APIs, asynchronous services, and ensuring high availability performance.
- Proficient in SQL databases including PostgreSQL, MySQL, and NoSQL databases such as Cassandra, Redis, MongoDB.
- Experience implementing message queuing systems such as Kafka, supporting event-driven and real-time data processing pipelines.
- Skilled in containerization and orchestration using Docker and Kubernetes, enabling cloud-native deployments across multiple environments.
- Designed and managed CI/CD pipelines using GitHub Actions, Terraform, Helm, for automated, secure, and repeatable deployments.
- Applied test-driven development (TDD) and automated testing frameworks like PyTest to ensure robust and reliable backend code.
- Strong foundation in object-oriented design (OOD), data structures, algorithms, and software engineering best practices.
- Worked in Linux environments with version control systems Git, GitLab, and collaborative development workflows efficiently.
- Experienced in cloud platforms including AWS, GCP, Azure, integrating services like Lambda, S3, Cloud Functions, Step Functions.
- Conducted performance optimization, load testing, stress testing, and analysis to support high-volume enterprise applications.
- Collaborated with cross-functional teams in Agile environments, mentoring junior developers, conducting reviews, and delivering features.
- Committed to writing clean, maintainable, efficient, and secure code while ensuring compliance with industry security standards.

TECHNICAL SKILLS:

Programming & Scripting	Python, SQL, Bash, PowerShell, Java
Frameworks & Libraries	Pandas, NumPy, Dask, PySpark, SQLAlchemy, Spring Boot,Pydantic, FastAPI, Flask, Django
AI/ML	Scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, Keras, PyTorch, TorchServe, NVIDIA Triton,YOLO, Faster R-CNN, SSD
NLP & GenAI	Hugging Face Transformers, spaCy, NLTK, LangChain, LangGraph, LlamaIndex, OpenAI SDK, Azure OpenAI, Cohere, SentenceTransformers
Vector & RAG	FAISS, Pinecone, Chroma, Redis Vector, Rerankers, Prompt Templates, Tool Use, Conversation Memory
MLOps & Data Engineering	MLflow, DVC, Kubeflow, Apache Airflow, Feast, Evidently AI, Great Expectations, Argo CD, Helm, Apache Spark, AWS Glue, dbt, Kafka, Kinesis, Pub/Sub, Dataflow
Databases & Warehouses	PostgreSQL, SQL Server, Oracle, MySQL, Redis, MongoDB, Snowflake, Redshift, Delta Lake , Cassandra
Cloud & DevOps	AWS (EKS, Lambda, Glue, Redshift), GCP (GKE, Cloud Run, Vertex AI, BigQuery), Azure, Kubernetes, Docker, Terraform, CI/CD pipelines
APIs, Messaging & Caching	FastAPI, Flask, Django REST, gRPC, OpenAPI/Swagger, OAuth2, JWT, Kafka, RabbitMQ, Redis
Security, Monitoring & Testing	IAM, VPC, Secrets Manager, HashiCorp Vault, KMS, CloudTrail, HIPAA, SOC 2, Prometheus, Grafana, CloudWatch
Visualization & Tools	Matplotlib, Seaborn, Plotly, Dash, Tableau, Looker, JupyterLab, PyCharm, VS Code, Jira, Confluence, Agile, Scrum

PROFESSIONAL EXPERIENCE:

Cardinal Health - Dublin, Ohio
Senior Python and ML Engineer
Responsibilities:

Jan 2024 to Present

- Designed and deployed highly scalable backend services for clinical and operational healthcare systems using Django and Flask, handling patient, pharmacy, and claims data securely.
- Developed RESTful APIs and asynchronous services ensuring high availability, low-latency processing, and reliability for clinical and operational healthcare data workflows.
- Implemented event-driven pipelines using AWS Lambda, Step Functions, S3, CloudWatch, and integrated Kafka for real-time clinical and operational data processing.
- Built and optimized end-to-end ETL pipelines with Pandas, PySpark, AWS Glue, and Databricks, improving processing efficiency by 40% for patient, pharmacy, and claims data.
- Automated PHI/PII detection and HIPAA-compliant redaction using spaCy and Microsoft Presidio before storing sensitive data in S3, Redshift, and Databricks.
- Developed monitoring and observability frameworks with CloudWatch, Prometheus, and Grafana to track API, ETL, and data pipeline performance.
- Designed modular, object-oriented backend components with caching, authentication, and role-based access control to maintain secure healthcare systems.
- Migrated legacy ETL pipelines to Databricks Lakehouse and AWS using Terraform and Helm, implementing infrastructure-as-code for repeatable deployments.
- Built CI/CD pipelines with GitHub Actions, Docker, Helm, and Terraform; applied TDD and automated testing frameworks to ensure robust releases.
- Implemented blue-green and canary deployments for APIs and ETL workflows to minimize downtime and ensure seamless healthcare service delivery.
- Conducted load testing, stress testing, and performance tuning using Locust and JMeter for high-volume healthcare applications.
- Developed reusable Python modules for ETL, API, and backend workflows to standardize healthcare data processing across engineering teams.
- Ensured cross-platform Linux deployments and container orchestration using Docker and Kubernetes for secure and scalable healthcare environments.
- Collaborated with frontend developers, PMs, and DevOps teams to deliver HIPAA-compliant, secure, and scalable software solutions.
- Applied data validation and schema enforcement for healthcare datasets using Great Expectations and Pandera, ensuring high-quality, consistent data.
- Optimized data pipelines and Python workflows for high-volume healthcare transactions, improving throughput and reliability.
- Integrated messaging systems (Kafka, Redis) to enable real-time event-driven processing of healthcare and pharmacy workflows.
- Provided mentorship and code reviews to junior engineers, promoting best practices in Python development, healthcare data compliance, and secure coding standards.

Environment: Python (Django, Flask), PostgreSQL, MongoDB, Redis, Pandas, PySpark, Databricks, AWS (Lambda, S3, Glue, Step Functions, Redshift, CloudWatch), Kafka, Docker, Kubernetes, Terraform, Helm, GitHub Actions, Prometheus, Grafana, Plotly Dash, Streamlit, Linux, CI/CD pipelines, PyTest, Object-Oriented Design, REST APIs, Event-Driven Architecture, HIPAA Compliance, High Availability Systems.

Mastercard - Purchase, New York
Senior Python and ML Engineer

May 2021 to Dec 2023

Responsibilities:

- Developed and maintained high-performance Python REST APIs using Flask and Django, supporting batch and real-time transaction processing for payments and financial operations
- Built and automated end-to-end ETL pipelines with Python, Airflow, and Azure Data Factory, improving efficiency and enabling analytics- and compliance-ready datasets.
- Designed data preprocessing, feature engineering, and transformation scripts in Python, Pandas, and PySpark to support fraud analytics, scoring, and ML workflows.
- Implemented real-time streaming pipelines with Python, Kafka, and Azure Event Hubs to ingest and process high-volume payment, transaction, and event data.
- Developed a centralized feature store using Python and Redis to provide standardized, reusable features for fraud detection, scoring, and transaction analytics workflows.
- Created interactive dashboards with Plotly Dash and Python to monitor ETL pipelines, API throughput, latency, and payment platform performance metrics.
- Applied automated data validation frameworks using Great Expectations and Pandera, ensuring consistent and accurate datasets for reporting and PCI compliance.
- Containerized Python applications with Docker and deployed on Azure Kubernetes Service (AKS) using Helm for scalable, fault-tolerant payment microservices.
- Built CI/CD pipelines using GitLab CI, Jenkins, Terraform, and MLflow for automated testing, deployment, and version control of APIs and ETL workflows.
- Automated cloud infrastructure provisioning using Python SDKs and Terraform for Azure resources, ensuring reproducible and secure cloud environments.
- Implemented shadow and canary deployments to safely release APIs and ETL services, minimizing downtime and ensuring stable production rollouts.
- Optimized Python and Spark workloads for large-scale data processing, improving throughput for transaction logs, audit data, and event pipelines.

- Developed Python utilities for logging, monitoring, and alerting, integrated with Prometheus, Grafana, and Azure Monitor for full-stack observability.
- Secured Python applications using Azure Key Vault, Kubernetes Secrets, and IAM policies, ensuring adherence to PCI-DSS and enterprise security guidelines.
- Built reusable Python modules and libraries to standardize backend workflows and accelerate development of payment services and analytics pipelines.
- Collaborated with DevOps, QA, and data teams to implement best practices in Python automation, testing, CI/CD, and high-availability system design.
- Worked extensively with SQL (PostgreSQL, MySQL) and NoSQL (Cassandra, MongoDB, Redis) for transactional storage, caching, and data processing support.
- Ensured cross-platform compatibility and Linux deployments for backend services and containerized applications while mentoring junior engineers on best practices.

Environment: Python (Flask, Django), Pandas, PySpark, Airflow, Azure Data Factory & Data Lake, Synapse, AKS, Kafka, Redis, Docker, Helm, Terraform, Jenkins, GitLab CI, Plotly Dash, Prometheus, Grafana, Azure Key Vault, PostgreSQL, MySQL, Cassandra, MongoDB, Redis.

Delta Airlines - Atlanta, Georgia

Jan 2019 to April 2021

Senior Python Developer

Responsibilities:

- Developed high-performance backend APIs and microservices using Python, FastAPI, and Django REST Framework to manage large-scale aviation and booking data.
- Engineered data ingestion and transformation workflows in PySpark and AWS Glue, processing millions of flight and payment records stored in S3 and Redshift.
- Designed real-time ETL pipelines with Snowpipe, Streams, and Tasks for continuous synchronization between operational systems and Snowflake.
- Built event-driven and serverless solutions using AWS Lambda, Step Functions, and SQS to automate flight event processing and reporting triggers.
- Created comprehensive monitoring dashboards using CloudWatch, Prometheus, and Grafana to visualize system performance and reduce downtime.
- Implemented CI/CD pipelines with Docker, Jenkins, and Helm, automating build, testing, and deployment workflows across AWS environments.
- Tuned Spark and SQL workloads on AWS EMR to reduce job execution times and improve overall resource efficiency.
- Enforced cloud security and access controls through IAM roles, VPC configurations, KMS encryption, and Secrets Manager policies.
- Designed Python-based logging and audit frameworks for proactive monitoring and exception tracking across distributed systems.
- Integrated data validation and schema enforcement using Pandera and custom scripts to ensure quality in production data pipelines.
- Developed streaming connectors between Kafka and S3 for near real-time flight event capture and processing.
- Deployed and scaled containerized microservices on AWS EKS using Terraform templates and Helm charts for automated provisioning.
- Collaborated with cross-functional teams to refine technical requirements, optimize ETL strategies, and ensure stable production releases.
- Conducted code reviews, testing, and performance tuning, improving API throughput and deployment reliability.
- Authored detailed cloud architecture and deployment documentation, ensuring consistency across teams and future scalability.

Environment: Python, FastAPI, Django REST, PySpark, AWS EMR, AWS Glue, AWS Lambda, S3, Redshift, Snowflake, API Gateway, Step Functions, SQS, CloudWatch, Prometheus, Grafana, Docker, Jenkins, Helm, Terraform, PostgreSQL, Pandera, Kafka, KMS, IAM, VPC, Secrets Manager, Jira, Confluence, Pandas, NumPy.

Blue Rose Technologies – Hyderabad, India

June 2016 to Sept 2018

Python Developer

Responsibilities:

- Designed and developed scalable backend business modules using Python and Flask, integrating SQL databases (PostgreSQL, MySQL) and external REST APIs for enterprise-grade applications.
- Created robust ETL and data migration scripts that cleaned, transformed, and loaded legacy data into relational databases, improving system accuracy and reducing manual errors.
- Built Python automation utilities for data validation, report generation, file transformation, and monitoring of recurring operational tasks, enabling faster decision-making.
- Implemented database interaction layers using SQLAlchemy, optimized SQL joins, indexes, and stored procedures to ensure high-performance and scalable solutions.
- Collaborated with front-end teams to integrate APIs with Angular-based UIs, ensuring smooth and efficient data exchange and minimal latency between client and server layers.
- Developed scheduled reporting jobs exporting Excel and PDF summaries for business teams, supporting KPI tracking and operational analytics.
- Introduced comprehensive error handling, logging, and input validation mechanisms to improve API robustness, reliability, and maintainability.

- Deployed services using Docker containers and managed CI/CD pipelines with Jenkins, ensuring repeatable, reliable deployments across development and QA environments.
- Wrote unit and integration tests using PyTest and Postman, improving API reliability and preventing regression issues.
- Worked closely with QA and DevOps teams to troubleshoot production issues, optimize release turnaround, and maintain stable operations.
- Participated in Agile sprints, code reviews, and cross-functional collaboration to deliver maintainable, reusable, and secure Python components.
- Supported early-stage AI/ML initiatives by preparing datasets, building preprocessing scripts, and assisting in proof-of-concept experiments for analytics and machine learning workflows.

Environment: Python, Flask, PostgreSQL, MySQL, SQLAlchemy, REST APIs, Docker, Jenkins, PyTest, Postman, Agile, ETL, Data Pipelines

Bachelor's in Information Technology – ANU, Hyd, India (2017)